Kushagra Gupta

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EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY, DELHI

Bachelor's and Master's of Technology in Computer Science and Engineering

November 2021— July 2026 (expected) | India

CGPA : 9.517/10 | Department Rank 1

Relevant Coursework: Algorithms, OS, Compiler Design, Parallel & Distributed Programming, Computer Architecture, AI, ML, Cryptography, PL, Network & System Security, DBMS, Logic, Formal Methods, Algorithmic Graph Theory, Probability, Econometrics

EXPERIENCE

MPI-SWS, SAARBRÜCKEN, GERMANY | Hybrid Framework for Secure MPC Protocols (May 2024 - Present)

- Working with Prof. Deepak Garg on building a **novel hybrid multi-protocol** framework for Secure **Multi-Party Computation.**
- Implemented the **C++** framework on top of MP-**SPDZ** & extended it to support transfer between arithmetic & binary protocols.

JANE STREET SEE : IIT EDITION, HONG KONG | Trading & SWE Programme (Dec 2023)

- Learnt **OCaml, trading research and statistical modeling,** and the tools and libraries integral to Jane Street's everyday work.
- Developed the backend and UI for a computer game in **OCaml** and **Dune,** using the acquired skills and technologies.

INRIA SOPHIA ANTIPOLIS, FRANCE - TITANE TEAM | Parsimonious Alpha Wrapping (May 2023 - July 2023)

- Worked with Prof. Pierre Alliez on improving the **Alpha Wrapping with an offset algorithm** in the open-source **CGAL** Library.
- Implemented a parsimonious approach to **reduce** algorithm complexity by up to **50%** for efficiently generating mesh wraps.

VDO.AI, IIM AHMEDABAD & CORNELL | Video Long-Short Ads (Aug 2022 - Jan 2023)

- Worked with Prof. Anuj Kapoor, IIMA and Prof. Omid Rafieian, Cornell on the **Data Science** Project using **Pandas & Sklearn.**
- Improved **ad-revenue** by analyzing data for Skippable, Non-Skippable & No-Ads Videos based on Ad-Video quartile completion.

KEY PROJECTS

ADVANCED MEMORY MANAGEMENT AND SCHEDULING IN XV6 | Prof. Abhilash Jindal (Jan 2024 - May 2024)

- Implemented demand paging along with the **COW (Copy on Write)** mechanism in **C** for the **xv6 Operating System.**
- Designed a **priority scheduler** to prioritize foreground processes while preventing starvation and added **undo logging** in xv6.

RELIABLE TRANSFER OVER UDP | DB362 DBMS | Prof. Kaustubh | Prof. Aaditeshwar (Jan 2024 - May 2024)

- Developed a **UDP-based** protocol for reliable data transfer with TCP-like features & **congestion control,** optimized with **AIMD.**
- Implemented our DBMS using **Java,** Apache **Calcite,** focusing on B+ Tree Indexing, Query Parsing, Execution & Optimization.

LLVM CODE GENERATION FOR C | PL(0) INTERPRETER | Prof. Sorav Bansal | Prof. SAK (Jan 2024 - May 2024)

- Developed an **AST & LLVM** bitcode generator for **C** programs, with support for complex constructs and **local optimisations.**
- Implemented a **PL(0)** interpreter with **static** type checking, AST generation, and execution using **SML, ML-Lex,** and **ML-Yacc** .

ROLLERBALL CHESS PLAYING BOT | AR-MAXILLOFACIAL | Prof. Mausam | Prof. Rahul Narian (Aug 2023 - Nov 2023)

- Implemented adversarial **search** algorithms with **minimax** tree search, optimized with α-β pruning and heuristic eval functions.
- Developed an AR app in **Unity** for maxillofacial surgery with **U-Net** Transformers for mesh extraction & AMASSS segmentation.

ACHIEVEMENTS

2024	**DAAD-WISE** & PLMW Scholar	Scholarships awarded for internship at MPI-SWS (by **DAAD**) and for **PLMW@PLDI'24**
2023	CMMRS Participant	**Cornell, Maryland, Max Planck** Pre-doctoral Research School (Top **90** students worldwide)
2023	**Charpak Lab** Scholar	Scholarship awarded by **IFI - Embassy of France** for internship at **INRIA Sophia Antipolis**
2023	**UPE/ACM** Scholarship Awardee	Scholarship awarded by **UPE & ACM** for academic excellence to **Top 6** students worldwide
2023	IITD Semester Merit Award	Conferred for being in the **Top 7%** students in 1st, 2nd, 3rd, 4th & 5th semester
2022	Resource Person, APhO '22	Responsible for evaluating 22nd **Asian Physics Olympiad** hosted by IAPT, Dehradun, India
2022	Reliance Foundation Scholar	AI Fellowship awarded by **Reliance** Foundation to brightest 60 students across India
2021	JEE(A) & JEE(M) **AIR 109 & 72**	Secured **Rank 109 & 72** in JEE Advanced and JEE Main amongst 1 Million Candidates
2021	**MOOC-INMO Qualifier**	**Math Olympiad Orientation Camp** held in lieu of **IMOTC** , for the 62nd **IMO Russia**
2021	KVPY & NTSE Fellow	Awarded KVPY (AIR **35** (SX) & **158** (SA)) & NTSE Fellowships by IISc, Bangalore & NCERT
2021	NSEP, NSEC, NSEA **Top 1** %	**National Top 1** % amongst Top **209, 179 & 82** candidates in IPhO, IChO & IOAA Stage-1
2021	RMO, NSEC, NSEA Qualifier	Qualified **NSEC, NSEA, RMO** 2021, **NSEA** 2020 & **RMO** 2019 conducted by **HBCSE**

EXTRACURRICULARS

UG **Teaching Assistant** - 2401-**COL215** — Taught, graded, and conducted **VHDL** labs for the course Digital Logic and System Design

Coordinator, iGEM, ACES-ACM, BlocSoc — **Secretary,** ACM Student Chapter and **Coordinator,** iGEM and Blockchain Society, IITD

Tower Research Data Challenge '23 & '24 — TRC Data Challenge **Finalist** | 1st **&** 3rd Position, Dalal Street & Abstractify, Tryst'23, IITD

1st **Position, Techfest'22, IITB** — In Stress Prediction, Techfest'22 | Qualified for IITB Winter School'22 - **ACM & Trust Lab**

BSW Student Mentorship '23 and '24 — Student Mentor for 1st Year CS5 students & academic mentor for **MTL100, Calculus**